

Table 2. Response quality and preservation of general QA performance

| Method                                                              | Qwen2.5 7B        |                    | GPT-6 Luna        |                   | Gemma 4 12B       |                    | Qwen3.8 27B OFF   |                   | Qwen3.8 27B ON    |                   |
|---------------------------------------------------------------------|-------------------|--------------------|-------------------|-------------------|-------------------|--------------------|-------------------|-------------------|-------------------|-------------------|
|                                                                     | FPQ ↑             | NFP / TPQ ↑        | FPQ ↑             | NFP / TPQ ↑       | FPQ ↑             | NFP / TPQ ↑        | FPQ ↑             | NFP / TPQ ↑       | FPQ ↑             | NFP / TPQ ↑       |
| Cancer-Myth                                                         |                   |                    |                   |                   |                   |                    |                   |                   |                   |                   |
| Plain                                                               | 2.4 <sup>H</sup>  | 100.0 <sup>H</sup> | 54.5 <sup>L</sup> | 91.0 <sup>L</sup> | 4.3 <sup>H</sup>  | 100.0 <sup>H</sup> | 49.7 <sup>H</sup> | 79.9 <sup>H</sup> | 40.7 <sup>H</sup> | 88.6 <sup>H</sup> |
| CoT                                                                 | 0.7 <sup>H</sup>  | 99.3 <sup>H</sup>  | —                 | —                 | 3.4 <sup>H</sup>  | 99.3 <sup>H</sup>  | 54.7 <sup>H</sup> | 71.1 <sup>H</sup> | 43.1 <sup>H</sup> | 87.9 <sup>H</sup> |
| Balanced instruction                                                | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| Review CoT (2-step)                                                 | 10.5 <sup>H</sup> | 94.6 <sup>H</sup>  | —                 | —                 | 41.2 <sup>H</sup> | 78.5 <sup>H</sup>  | 61.4 <sup>H</sup> | 55.0 <sup>H</sup> | 66.7 <sup>H</sup> | 57.0 <sup>H</sup> |
| Always correct                                                      | 56.9 <sup>H</sup> | 21.5 <sup>H</sup>  | —                 | —                 | 72.0 <sup>H</sup> | 2.0 <sup>H</sup>   | 71.0 <sup>H</sup> | 28.9 <sup>H</sup> | 83.4 <sup>H</sup> | 5.4 <sup>H</sup>  |
| PreWoMe-style                                                       | —                 | —                  | 88.9 <sup>L</sup> | 59.0 <sup>L</sup> | —                 | —                  | —                 | —                 | —                 | —                 |
| Extract + Verify                                                    | —                 | —                  | 87.9 <sup>L</sup> | 48.0 <sup>L</sup> | —                 | —                  | —                 | —                 | —                 | —                 |
| Extract+Verify (scope)                                              | 5.7 <sup>H</sup>  | 99.3 <sup>H</sup>  | n/a               | n/a               | n/a               | n/a                | n/a               | n/a               | n/a               | n/a               |
| Self-gated FP identification                                        | 7.0 <sup>H</sup>  | 89.9 <sup>H</sup>  | n/a               | n/a               | n/a               | n/a                | n/a               | n/a               | n/a               | n/a               |
| Direct-gated response                                               | 6.9 <sup>H</sup>  | 90.6 <sup>H</sup>  | —                 | —                 | 37.2 <sup>H</sup> | 83.9 <sup>H</sup>  | 53.0 <sup>H</sup> | 72.5 <sup>H</sup> | 53.5 <sup>H</sup> | 75.8 <sup>H</sup> |
| CoT-gated response                                                  | 21.3 <sup>H</sup> | 85.9 <sup>H</sup>  | —                 | —                 | 43.4 <sup>H</sup> | 67.1 <sup>H</sup>  | 62.1 <sup>H</sup> | 59.7 <sup>H</sup> | 62.1 <sup>H</sup> | 60.4 <sup>H</sup> |
| TF-IDF text gate                                                    | 50.7 <sup>H</sup> | 60.4 <sup>H</sup>  | —                 | —                 | 63.8 <sup>H</sup> | 50.3 <sup>H</sup>  | 68.3 <sup>H</sup> | 49.0 <sup>H</sup> | 78.2 <sup>H</sup> | 48.3 <sup>H</sup> |
| Probe-gated response                                                | 44.8 <sup>H</sup> | 64.4 <sup>H</sup>  | n/a               | n/a               | —                 | —                  | —                 | —                 | —                 | —                 |
| GEPA                                                                | —                 | —                  | 78.8 <sup>L</sup> | 72.0 <sup>L</sup> | —                 | —                  | —                 | —                 | —                 | —                 |
| Ours                                                                | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| CREPE                                                               |                   |                    |                   |                   |                   |                    |                   |                   |                   |                   |
| Plain                                                               | —                 | —                  | 72.7 <sup>R</sup> | 98.9 <sup>R</sup> | —                 | —                  | —                 | —                 | —                 | —                 |
| CoT                                                                 | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| Balanced instruction                                                | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| Review CoT (2-step)                                                 | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| Always correct                                                      | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| PreWoMe-style                                                       | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| Extract + Verify                                                    | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| Extract+Verify (scope)                                              | —                 | —                  | n/a               | n/a               | n/a               | n/a                | n/a               | n/a               | n/a               | n/a               |
| Self-gated FP identification                                        | —                 | —                  | n/a               | n/a               | n/a               | n/a                | n/a               | n/a               | n/a               | n/a               |
| Direct-gated response                                               | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| CoT-gated response                                                  | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| TF-IDF text gate                                                    | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| Probe-gated response                                                | —                 | —                  | n/a               | n/a               | —                 | —                  | —                 | —                 | —                 | —                 |
| GEPA                                                                | —                 | —                  | 74.7 <sup>R</sup> | 98.0 <sup>R</sup> | —                 | —                  | —                 | —                 | —                 | —                 |
| Ours                                                                | —                 | —                  | —                 | —                 | —                 | —                  | —                 | —                 | —                 | —                 |
| Medical / general QA: accuracy (%) or GSM8K EM; all methods pending |                   |                    |                   |                   |                   |                    |                   |                   |                   |                   |
| MedQA (all methods pending)                                         | —                 |                    | —                 |                   | —                 |                    | —                 |                   | —                 |                   |
| PubMedQA (all methods pending)                                      | —                 |                    | —                 |                   | —                 |                    | —                 |                   | —                 |                   |
| Medbullets (all methods pending)                                    | —                 |                    | —                 |                   | —                 |                    | —                 |                   | —                 |                   |
| MMLU (all methods pending)                                          | —                 |                    | —                 |                   | —                 |                    | —                 |                   | —                 |                   |
| GSM8K (all methods pending)                                         | —                 |                    | —                 |                   | —                 |                    | —                 |                   | —                 |                   |

Stored results; —: not measured/comparable; n/a: outside the defined implementation. Percentages, one run.  
H: historical full-corpus Cancer-Myth; valid denominators and protocol differences in ledger.  
FPQ / NFP / TPQ: Well ≥ 4. MQ / PQ / MB: MedQA / PubMedQA / Medbullets.  
L: Luna matched 99 FPQ / 100 NFP, exposed test; R: CREPE 751 FPQ / 2253 TPQ.  
Gate rows route saved Plain/correction answers; not exact Two Axes reproduction.  
Ours is not fixed to SAE. Source counts and comparison limits: result\_sources.md.  
QA rows with no measurements are compressed across methods; full planned cells remain in CSV.